

SOLUTION OF HIGH ASPECT RATIO TSV PVD IN ADVANCED PACKAGING

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ABSTRACT

With the Moore's law is approaching physical limit, and the big data and artificial intelligence rapidly developing, more and more attention is focused on the advanced packaging. The TSV interconnect is one of the most important process in the advanced packaging. Cost down and performance improvement of TSV tool is very critical and important to the advanced packaging industry.

AMEC has developed Primo InnoFill PVD system targeting at high aspect ratio TSV for the advanced packaging. The PVD system integrates several unique technology, including low cost and high throughput dual stations design, unique side electromagnet assembly design, adding high frequency RF to DC power to target and so on. These unique designs of the Primo InnoFill PVD system deliver good RS uniformity, high step coverage with significantly lower cost. In this paper, we introduce the details of AMEC Primo InnoFill PVD design and the performance.

Keywords—advanced packaging; TSV; PVD; low cost; step coverage; AMEC

INTRODUCTION

As the big data and artificial intelligence rapidly developing, the demand for higher input/output counts is increasing. While the Moore's law is approaching physical limit, advanced packaging has been developed, which showing significant growth recently. 2.5D/3D integration, as an innovative interconnect technology, is becoming the trend for enhancing I/O bandwidth in advanced packaging processing. Usually, TSV is used as a vertical interconnect in 2.5D/3D integration technology, enabling vertical electrical connections between silicon wafers or dies. This technology interconnects multiple chips, memory, sensors, and other modules, significantly enhancing data transfer rates by allowing direct vertical signal transmission and minimizing parasitic effects like resistive-capacitive delay. In addition to enhancing electrical performance, TSVs also facilitate higher density of inputs and outputs by being distributed across the entire surface area of a chip, rather than being restricted to the chip's perimeter. Furthermore, TSVs enhance thermal management within the chip. They facilitate the vertical stacking of multiple semiconductor chips, thereby allowing for the packaging of microelectronic devices

with increased functionality per unit volume.^[1] Therefore, TSVs are particularly well-suited for high-speed applications, including data centers, servers, graphics processing units, artificial intelligence processors, and wireless communication devices.

In the recent decades, Low aspect ratio (LAR, AR <10) TSV technology has successfully transitioned into high-volume manufacturing for CMOS imaging sensors, SiGe power amplifiers, and advanced packaging products. Conversely, the high aspect ratio (HAR) TSVs for 2.5D/3D packaging remain in the early stages of volume manufacturing. This predicament is attributed to the growing challenges associated with the rising aspect ratio.^[2] The HAR TSV for 2.5D/3D packaging gives many challenges to the semiconductor industry. PVD process, as one of the most important technologies to provide the barrier and seed for TSV, faces many challenges, such as deposition uniformity and step coverage. In order to meet the requirements of HAR TSV, innovation of the PVD chamber design have to be considered. Meanwhile, cost performance is also very important for advanced packaging technology to be widely adopted rapidly.

The below section will introduce the details of AMEC Primo InnoFill PVD system to meet the requirements.

DISCUSSION

The Primo InnoFill PVD system is a cluster PVD system, including transfer module, degas module, pre-clean module and PVD module. This PVD system has not only many unique designs to improve the process performance, but also lower cost to meet the requirement of industry.

The Primo InnoFill PVD system can configure one stage transfer module, with eight process stations installed. The transfer module has two degas stations, two pre-clean stations and four PVD stations, as illustrated in figure 1. With this system layout, more process stations can be integrated compared to conventional PVD system with same area. The Primo InnoFill PVD system is also optimized to achieve maximal throughput. In another words, the Primo InnoFill PVD system is designed to achieve optimal throughput and reduce the cost.

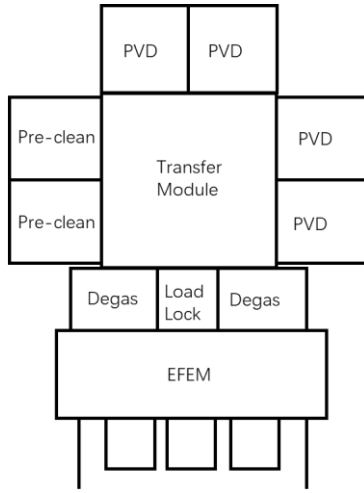


Figure 1: The Primo InnoFill PVD system layout with up to eight process stations.

The Primo InnoFill PVD system adopts dual mode degas, shown in figure 2. Both lamps and resistance heater can supply high temperature for the wafer. The degas module has two stations in one chamber. The two stations can share one high vacuum system with high pump speed and one gas supply system. The share parts can obviously reduce the cost of the degas module. Also the temperature of wafer can be controlled at a good temperature uniformity, $\pm 5^\circ\text{C}$ at 300°C .

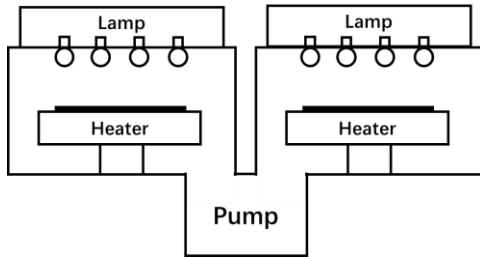


Figure 2: The Degas module with two stations in one chamber.

The Primo InnoFill PVD system adopts ICP pre-clean, shown in figure 3. The pre-clean inherits from the AMEC's unique twin chamber design, is production proven in mass production. The pre-clean adopts several unique designs. First is the dual coil design, which is a unique patented design of AMEC's ICP etch product. The dual coil on the top of the chamber can tune the outer and inner plasma density on the wafer surface in the chamber. Secondly, the pre-clean develops dual zone ESC, which can control the temperature of the outer and inner part in the wafer, therefore to control the etch amount in the outer and inner part of the wafer. The dual coil design and the two zone ESC can easily control the etch amount and tune the etch uniformity at a good condition. The pre-clean module also adopts two stations in one chamber. This design can share one vacuum system and one gas supply

system. So the pre-clean can not only effectively supply the tuning function of etch uniformity, but also obviously reduce the cost of the module. Figure 4 is the pre-clean performance of etching $50\text{ \AA}$ SiO_2 , which is at a stability condition with etch amount average $50.7\text{ \AA}$ and uniformity $<3\%$ (1σ).

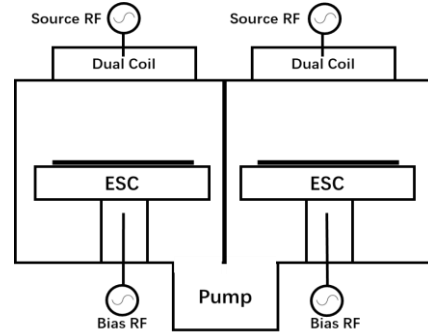


Figure 3: The pre-clean module with two stations in one chamber.

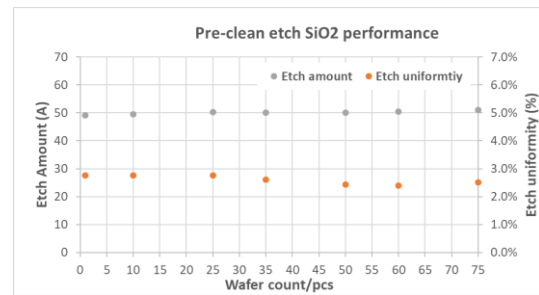


Figure 4: The pre-clean performance with etch amount $50\text{ \AA}$ SiO_2

The Primo InnoFill PVD system adopts PVD module with several special unique designs to reduce the cost and improve the process performance of PVD barrier and seed. The PVD module adopts the two stations design in one chamber. The design can also obviously reduce cost of the PVD module to meet the cost target, shown in figure 5.

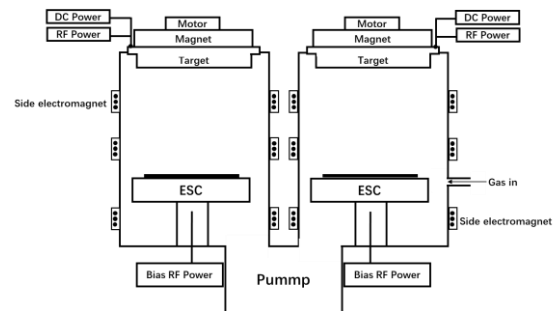


Figure 5: The PVD module with two stations in one chamber.

For HAR TSV, the high step coverage of barrier and seed is the most important and challenging thing for PVD process performance, which has a direct influence on the

next electroplate process and may impact the product yield. Several unique designs have been developed in the PVD module to achieve good step coverage of barrier and seed.

The first unique design is the side electromagnet assembly. The PVD module adopts three groups of side electromagnets. Two groups of side electromagnet are positioned above the wafer and one group is below the wafer. The group below the wafer can effectively tune the magnetic field in the chamber by cooperating with two other groups. The three groups of side electromagnet can have a wide process window of tuning the strength and direction of magnetic field. Because of the wide process window, the three groups of side electromagnet can flexibly tune the PVD process, including step coverage and RS uniformity. The figure 6 is the PVD Cu RS performance, which is at a good stability condition with RS average 43mΩ/sq and uniformity <2.5% (1σ).

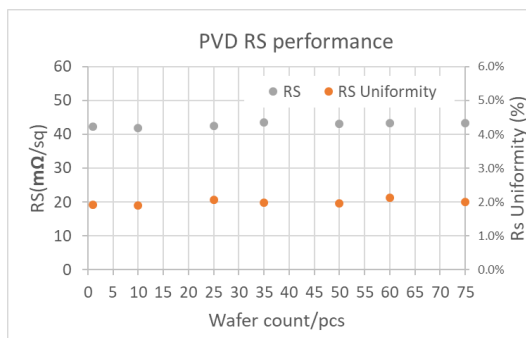


Figure 6: The PVD Cu RS performance.

The second unique design is the power loaded to the metal target. The high ionization rate of metal can improve the step coverage of HAR TSV. In the PVD module, both DC power and high frequency RF power can be loaded to the metal target. Two types of power together loaded to the metal target can obviously increase the ionization rate of metal. Therefore, plasma with high metal ionization rate can be effectively achieved, thus improving the step coverage.

The third unique design is the bias RF power loaded to the ESC. The bias RF power adopts the pulsing mode instead of continuous wave mode. In the pulsing RF mode, the RF power has 2 levels: the low-level RF power and the high-level RF power. The low-level RF power will deliver low energy ions, so that less re-sputtering occurs, while re-sputtering is enhanced at high energy ions with high RF power. By tuning duty cycle ratio of RF pulsing, or the power on time ratio of high RF power and low RF power, we can precisely control re-sputtering amount to sidewall, both at the shoulder and the bottom of TSV, so that good step coverage can be achieved.

We compare with the effect of two conditions. The first condition contains only DC loaded to metal target,

using two groups of side electromagnet above the wafer and bias continuous wave RF loaded. The second condition contains DC and RF loaded to metal target, using three groups of side electromagnet and appropriate bias pulsing RF power loaded. Compared with the effect at the first condition, the step coverage of barrier and seed for HAR TSV can be improved by 63 percent at the second condition.

SUMMARY

There are a lot of challenges to the PVD process in the high aspect ratio TSV, for instance, cost down, high throughput, good uniformity, high step coverage. AMEC Primo InnoFill PVD system can provide perfect solutions for all the challenges in the high aspect ratio TSV.

REFERENCES

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